

$$A = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix}$$

$$AA^T = \begin{pmatrix} 7 & -2 & -2 \\ -2 & 4 & -2 \\ -2 & -2 & 4 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 6 & -3 & -3 \\ 0 & -3 & 6 & -3 \\ 0 & -3 & -3 & 6 \end{pmatrix}$$

~~Matrice A je symetricka~~

vlastní čísla pro AA^T

$$\lambda_{1,2} = 9 \rightarrow \text{singulární čísla} \\ \lambda_3 = 3 \quad \sigma_1 = \sigma_2 = 3 \quad \sigma_3 = \sqrt{3}$$

vlastní vektor pro $\lambda = 3$

$$(AA^T - 3I)w = 0$$

$$\begin{pmatrix} 4 & -2 & -2 \\ -2 & 4 & -2 \\ -2 & -2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\left\| \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\| = \sqrt{3} \quad w_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\begin{cases} 2x - y - z = 0 \\ -x + 2y - z = 0 \\ -x - y + 2z = 0 \end{cases}$$

$$\left. \begin{matrix} x = y = z \\ \downarrow \end{matrix} \right\}$$

$$\text{ vlastní vektor: } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

vlastní vektor pro $\lambda = 9$

$$(AA^T - 9I)w = 0$$

$$\begin{pmatrix} -2 & -2 & -2 \\ -2 & -2 & -2 \\ -2 & -2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$x + y + z = 0 \rightarrow \text{vlastní vektor s normou } x + y + z = 0$$

$$\text{ vybereme 2 ortogonální vektory: } \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$$

$$\sqrt{1^2 + (-1)^2 + 0^2} = \sqrt{2}$$

$$\left\| \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\| = \sqrt{2}$$

$$\sqrt{1^2 + 1^2 + (-2)^2} = \sqrt{6}$$

$$\left\| \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \right\| = \sqrt{6}$$

$$\mu_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \quad \mu_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$\pi_i = \frac{1}{\sigma_i} A^T \mu_i$$

$$\pi_1 = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\pi_2 = \begin{pmatrix} 0 \\ \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{pmatrix}$$

$$\pi_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

duplikat nie rozróż. kolumn
jeden $\pi_4 = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$

$$V = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix}$$